

Inflation / Discounting and Present Value Factor (PVF) Calculation Methodology

14 January 2014

AMCOS Inflation Methodology

Inflation is a consistent rise in the costs (prices) of goods and services over time. AMCOS considers inflation by applying Inflation Factors to the Default Summary cost elements found in the Project Manager (PM). The Joint Inflation Calculator (JIC), as shown in Figure 1, is located on the OASA (FM&C) website, <http://asafm.army.mil/offices/CE/Rates.aspx?OfficeCode=1400>. It is the source for the eleven (11) inflation factors built into Project Manager (PM). See Attachment 1 for JIC Inflation Factors listed by pay plan for FY 2014 – 2060; Base Year =2014.

Note: All Pictures in this document can be enlarged by clicking on the Picture, holding down the control “Ctrl” Key on your keyboard, and using your scroll wheel on your mouse simultaneously.

Inflation Query Sheet

1. Select Service
⇒

NAVY

ARMY

Marine Corps

DoD Wide

ARMY

2. Select Appropriation/Cost Element from this List ⇒

Civilian Pay (CIVPAY) Appropriation
▼

3. Enter Base/Input Year(1985 - 2060)

2014

Generate Inflation Table

Go To SAR Calculator Worksheet

Optional - For Quick Look, complete steps A, B & C below

A. Select Inflation Type from List ⇒

Budget/Then-Year \$ to Budget/Then-Year \$
▼

B. Enter Output/Target Year ⇒

2060

C. Enter Starting Values in Input Column (blue cells) Below

Quick Look

Civilian Pay (CIVPAY) Appropriation			
9/23/2013 Print Quick Look	Input	Inflation Factor	Output/ Result
Years	2014	⇒	2060
Escalation Type	BY/TY\$	⇒	BY/TY\$
Enter starting values in the blue input cells. The inflation factor (based on your selected appropriation, year, and type) is applied and the results given in the output column	100.0	1.9519	195.2
	10,000.0	1.9519	19,519.1
	100,000.0	1.9519	195,191.2
	1,000.0	1.9519	1,951.9
	1,000.0	1.9519	1,951.9
	1,000.0	1.9519	1,951.9

Figure 1

Each component (Active, NG, & Reserves) have their own separate set of JIC “MPA”& “OMA” inflation factors. In addition, the Active component also applies the “MPA Non Pay” inflation factor to PCS Costs and is discussed in the next paragraph. Whereas, Civilian (GS, WG, WL, WS, SES, and) positions are only inflated by two inflation factors, “CivPay” or “OMA”. NOTE: The same “OMA” inflation factor is used for Active Military, Civilian, &

Private Labor Market (PLM) pay plans. For PLM positions, only “OMA” is applied due to most contracts are paid with “OMA” dollars.

The “MPA” inflation factor, for the specific components (Active, NG, Reserve), is applied to all MPA Appropriation (APPN) cost elements except Permanent Change of Station (PCS) related cost elements. In this case, the “MPA Non Pay” inflation factor is applied to PCS cost elements. It sometimes may also be used for Clothing and Allowances, but it is more important for PCS, as PCS is a much larger portion of Army money. When AMCOS LITE displays APPN = “OMA”, then the specific components (Active, NG, Reserve) “OMA” inflation factor is applied to any cost element with either APPN.

Note: The following two cost elements are not inflated in Project Manager, “Average Cost of Special Pay” and “Average Cost of MGIB-AD Army College Fund (ACF)”.

Refer to Attachment 2 to see a list of Inflation Factors mapped to DEFAULT Summary Cost Elements.

For Project Manager Purposes, we’ll refer to Fiscal Year (FY) costs generated in AMCOS LITE as “BASE YEAR = 2014” costs. PM multiplies BASE YEAR costs by the appropriate target years INFLATION FACTOR to generate your Future Start Year and/or your Future Year Cost Element costs across the entire duration of a project. For example, let’s say you set the Project Start Year = 2016, set the Duration = 5 years, and are costing a NG position with focus on the “Salary” cost element:

- 2016 NG Salary = (2014 BASE YEAR Salary) * (2016 MPA-NG Inflation Factor).
- 2017 NG Salary = (2014 BASE YEAR Salary) * (2017 MPA-NG Inflation Factor).
- The Inflation Process repeats until the specified duration completes, in this case, 5 years.

AMCOS DISCOUNTING and Present Value Factor (PVF)

Discounting and Present Value Information:

Most cost comparison techniques take into consideration the time value of money, that is, a dollar today is worth some amount less in the future. For comparison purposes, future expenditures, occurring at different points in time, must be adjusted to a common point in time. This adjustment to a common point in time is called discounting or present value analysis. Discount factors can be calculated once an interest rate and period of analysis is determined. These discount factors indicate the present value (today’s value) of a dollar expended at mid-year or the end of each respective year (AMCOS applies mid-year discounting). Therefore, discounting converts various cash flows occurring over time to equivalent amounts at a common point in time for purposes of comparison. This accounts for differing interest costs and the time value of money resulting from expenditures over varying time periods.

Office of Management and Budget (OMB) Circular A-94 and Department of Defense Instruction (DoDI) 7041.3 require the use of a discount rate based on the Treasury Department cost of borrowing funds. This discount rate should be used in evaluating the measurable costs and/or benefits of programs or projects when they are distributed over time. The prescribed rate will vary dependent on the length of the period of analysis and on whether the costs and benefits are measured in constant or current dollars. A discount rate that reflects expected inflation should be used to discount costs and benefits expressed in current dollars, as is the case in

AMCOS. Conversely, a discount rate that has already been adjusted to eliminate the effect of expected inflation should be used to discount costs and benefits expressed in constant dollars.

The discount rate for use in DoD economic analyses is based on the U.S. Treasury Department's cost of borrowing funds and reflects the expected cost of borrowing for 3, 5, 7, 10, 20, and 30 year securities. Annually, the OMB issues an update of the borrowing rates based on the estimates provided in the President's budget submission. Typically, the update is issued in February and remains applicable until the next annual update.

AMCOS uses the six basic rates (3, 5, 7, 10, 20, and 30 year securities) and the user's Project Manager Duration input determines which rate applies.

AMCOS Discounting and Present Value REVIEW:

During March 2013, DASA-CE reviewed the AMCOS Discounting methodology and concluded the following:

1. The AMCOS Default Summary Military Rates are expressed in current dollars. There are MPA and OMA dollars in that summary. We do not convert our OMA costs to constant dollars. Therefore, the summary is expressed in current dollars.
2. The Default Summary Civilian costs use OMA for the government-funded costs and Civilian Pay for the rest of the cost factors, so they are also expressed in current dollars due to the OMA. We do not convert the government costs into constant dollars.
3. The Civilian rates mostly use OMA, as a result, they are expressed in current dollars. We do not convert the OMA costs into constant dollars. The non OMA costs do not require a conversion.
4. FINDING: Apply the nominal discount rates for AMCOS.

Discounting and Present Value GUIDANCE:

1. **DoD INSTRUCTION Number 7041.3** (Link: <http://www.dtic.mil/whs/directives/corres/pdf/704103p.pdf>)
November 7, 1995

SUBJECT: Economic Analysis for Decision Making

REF: Page 12

The proper discount rate to use depends on whether the costs and benefits are measured in current or constant dollars.

- If costs and benefits are expressed in constant dollars:
 - A real discount rate; i.e., a nominal rate that has been adjusted to exclude expected inflation, should be used to calculate a net present value.
- If costs and benefits are measured in current dollar:
 - Then a nominal discount rate (which implicitly includes inflation) should be used to calculate the net present value.

REF: E3.A3. ENCLOSURE 3, ATTACHMENT 3. DETERMINING THE DISCOUNT RATE FOR PERFORMING ECONOMIC ANALYSIS OF INVESTMENT ALTERNATIVES.

E3.A3.3.2. When costs and benefits occur in a steady stream, applying mid-year factors is best for the analysis.

2. **Department of the Army**
Economic Analysis Manual
U.S. Army Cost and Economic Analysis Center
February 2001
REF: Chapter 3, Page 21

(2) Current dollars. Current year dollars are expressed in the value of the year of in which a cost is expected to occur, and therefore reflect the effects of inflation. The term "current year" does not refer to the year in which the estimate is made or any other single year. It means that the amount is appropriate for the year in which the dollars are expected to be expended.

3. **M-13-04** (Link: <http://www.whitehouse.gov/sites/default/files/omb/memoranda/2013/m-13-04.pdf>)
Memorandum for Heads of Departments and Agencies
January 24, 2013
Subject: 2013 Discount Rates for OMB Circular No. A-94
REF: Appendix C

Nominal Discount Rates. A forecast of nominal or market interest rates for calendar year 2013 based on the economic assumptions for the 2014 Budget is presented below. These nominal rates are to be used for discounting nominal flows, which are often encountered in lease-purchase analysis.

Nominal Interest Rates on Treasury Notes and Bonds of Specified Maturities (in percent)

3-Year	5-Year	7-Year	10-Year	20-Year	30-Year
0.5	1.1	1.5	2.0	2.7	3.0

AMCOS Project Manager Discounting and Present Value Factor (PVF):

Project Manager automatically calculates the Discount Rate using the Present Value Factor (PVF) calculation which will be discussed momentarily. You may be thinking “we’re costing people (military or civilian); it’s not as though we’re costing buildings; do we really need to discount? The answer is yes, you need discounting even for people. Their benefits and other costs will be cheaper in the future than if we paid those benefits in today's dollars. What about projects that are only 1-2 years in duration? Again, the answer is yes because you need the ability to compare with other projects due to money being worth more today than in the future. You still need discounting.

There is no need to turn discounting on and off in Project Manager. Discounting is automatically calculated based on the duration of the project and displays in Project Manager’s output and exportable report. Output products provide both cost scenarios, Undiscounted (without discounting) and Discounted (with discounting), as well as displays a Discount Rates recap table applied by the model.

DASA-CE approved the following business rules for incorporation into the model:

1. Duration of the project will determine which rate to use.
2. Use Mid-Year Discount Factors. The mid-year exponent is expressed in the .5 power and will start during the first year with the 0.5 Exponent for the BASE YEAR. Example: 0.5, 1.5, 2.5, 3.5, 4.5, 5.5, etc.

3. Use the latest Nominal Rates cited by the annual DoD Memo regarding FY20xx Discount Rates for OMB Circular No. A-94.
4. Calculate PVF values based on the PVF formula.
5. To get the Discounted Cost, multiply the (PM Reports Undiscounted Cost Elements) by (PVF factor).

The Present Value Factor (PVF) Calculation:

1. **Determine which securities rate to use, 3-Year, 5 Year, etc.:** The model applies a separate Discount Rate for each year across the "Duration" of the project. The project's duration will determine which rate to use. For example, the user sets the Project's Duration = 6 years which falls in the 6 but less than 8 years range, therefore, the **7-Year Rate** nominal interest rate will be applied.

YEAR	LESS THAN	RATE Applied
1	4	3-Year Rate
4	6	5-Year Rate
6	8	7-Year Rate
8	11	10-Year Rate
11	21	20-Year Rate
21 or >	---	30-Year Rate

2. **Use Mid-Year Discount Factors:** For this **7-Year Rate** example, the exponent, "n" will equal:
 - Year 1 = 0.5
 - Year 2 = 1.5
 - Year 3 = 2.5
 - Year 4 = 3.5
 - Year 5 = 4.5
 - Year 6 = 5.5
 - Year 7 is NOT REQUIRED for a six year project; AMCOS just requires the correct nominal interest rate.
3. **Determine the Nominal Interest Rate:** Using the current dollars Nominal Interest Rates on Treasury Notes and Bonds of Specified Maturities (in percent), you can see the **7-Year Rate** is 1.5% as specified in the M-13-04 Memo extract below.

Nominal Interest Rates on Treasury Notes and Bonds of Specified Maturities (in percent)

3-Year	5-Year	7-Year	10-Year	20-Year	30-Year
0.5	1.1	1.5	2.0	2.7	3.0

4. **Calculate PVF Values:** The methodology for calculating the discount factors associated with the various discount rates is based on the following formula:

$$PVF_n = 1/(1+i)^{(n-.5)}$$

where PVF = the present value factor for year "n"
i = the discount rate

n = the project year

Continuing with our **7-Year Rate** example, the calculation of the mid-year discount factors for a **six** year project using current dollars is:

$$PVF_1 = 1/(1+.015)^{(0.5)} = 0.99258$$

$$PVF_2 = 1/(1+.015)^{(1.5)} = 0.97791$$

$$PVF_3 = 1/(1+.015)^{(2.5)} = 0.96346$$

$$PVF_4 = 1/(1+.015)^{(3.5)} = 0.94922$$

$$PVF_5 = 1/(1+.015)^{(4.5)} = 0.93520$$

$$PVF_6 = 1/(1+.015)^{(5.5)} = 0.92138$$

PVF₇ is not required.

The Figure 2 displays the PVF for all six Nominal Discount Rates. The numbers (0.5, 1.1, 1.5) are the discount rates.

Year	3-Year (0.5)	5-Year (1.1)	7-Year (1.5)	10-Year (2.0)	20-Year (2.7)	30-Year (3.0)
1	0.99751	0.99454	0.99258	0.99015	0.98677	0.98533
2	0.99255	0.98372	0.97791	0.97073	0.96083	0.95663
3	0.98761	0.97302	0.96346	0.9517	0.93556	0.92877
4		0.96243	0.94922	0.93304	0.91097	0.90172
5		0.95196	0.9352	0.91474	0.88702	0.87545
6			0.92138	0.89681	0.8637	0.84995
7			0.90776	0.87922	0.84099	0.8252
8				0.86198	0.81888	0.80116
9				0.84508	0.79735	0.77783
10				0.82851	0.77639	0.75517
11					0.75598	0.73318
12					0.73611	0.71182
13					0.71675	0.69109
14					0.69791	0.67096
15					0.67956	0.65142
16					0.6617	0.63245
17					0.6443	0.61402
18					0.62736	0.59614
19					0.61087	0.57878
20					0.59481	0.56192
21						0.54555
22						0.52966
23						0.51424
24						0.49926
25						0.48472
26						0.4706
27						0.45689
28						0.44358
29						0.43066
30						0.41812

Figure 2

5. **Calculate the Discounted Cost:** Multiply the (PM Reports Undiscounted Cost Elements) by (PVF factor). Finalizing our **7-Year Rate** example, the calculation of the mid-year discount factors for a **six** year project using current dollars is:

<u>PV FACTOR</u>	<u>Undiscounted Cost</u>	<u>Discounted Cost</u>
PVF ₁ = 0.99258	\$1,000	\$ 992.58
PVF ₂ = 0.97791	\$1,100	\$1,075.70
PVF ₃ = 0.96346	\$1,200	\$1,156.15
PVF ₄ = 0.94922	\$1,300	\$1,233.99
PVF ₅ = 0.93520	\$1,500	\$1,402.80
PVF ₆ = 0.92138	\$1,600	\$1,474.21
PVF ₇ is not required.		

Attachment 1

JIC Inflation Factors

Year	BUDGET YEAR INFLATION INDEX Active (AE, AO, & AWO)			
	Army MPA	Army MPA NonPay	Army OMA-MIL	DoD-OMA
2014	1.0000	1.0000	1.0000	1.0000
2015	1.0100	1.0190	1.0190	1.0190
2016	1.0201	1.0384	1.0384	1.0384
2017	1.0341	1.0581	1.0581	1.0581
2018	1.0597	1.0782	1.0782	1.0782
2019	1.0894	1.0987	1.0987	1.0987
2020	1.1199	1.1196	1.1196	1.1196
2021	1.1513	1.1408	1.1408	1.1408
2022	1.1835	1.1625	1.1625	1.1625
2023	1.2166	1.1846	1.1846	1.1846
2024	1.2507	1.2071	1.2071	1.2071
2025	1.2857	1.2300	1.2300	1.2300
2026	1.3217	1.2534	1.2534	1.2534
2027	1.3587	1.2772	1.2772	1.2772
2028	1.3968	1.3015	1.3015	1.3015
2029	1.4359	1.3262	1.3262	1.3262
2030	1.4761	1.3514	1.3514	1.3514
2031	1.5174	1.3771	1.3771	1.3771
2032	1.5599	1.4033	1.4033	1.4033
2033	1.6036	1.4299	1.4299	1.4299
2034	1.6485	1.4571	1.4571	1.4571
2035	1.6946	1.4848	1.4848	1.4848
2036	1.7421	1.5130	1.5130	1.5130
2037	1.7909	1.5417	1.5417	1.5417
2038	1.8410	1.5710	1.5710	1.5710
2039	1.8925	1.6009	1.6009	1.6009
2040	1.9455	1.6313	1.6313	1.6313
2041	2.0000	1.6623	1.6623	1.6623
2042	2.0560	1.6939	1.6939	1.6939
2043	2.1136	1.7260	1.7260	1.7260
2044	2.1728	1.7588	1.7588	1.7588
2045	2.2336	1.7923	1.7923	1.7923
2046	2.2961	1.8263	1.8263	1.8263
2047	2.3604	1.8610	1.8610	1.8610
2048	2.4265	1.8964	1.8964	1.8964
2049	2.4945	1.9324	1.9324	1.9324
2050	2.5643	1.9691	1.9691	1.9691
2051	2.6361	2.0065	2.0065	2.0065
2052	2.7099	2.0447	2.0447	2.0447
2053	2.7858	2.0835	2.0835	2.0835
2054	2.8638	2.1231	2.1231	2.1231
2055	2.9440	2.1634	2.1634	2.1634
2056	3.0264	2.2045	2.2045	2.2045
2057	3.1112	2.2464	2.2464	2.2464
2058	3.1983	2.2891	2.2891	2.2891
2059	3.2878	2.3326	2.3326	2.3326
2060	3.3799	2.3769	2.3769	2.3769

Year	BUDGET YEAR INFLATION INDEX NG (NE, NO, & NWO)	
	PA-NG	OM-NG
2014	1.0000	1.0000
2015	1.0105	1.0190
2016	1.0211	1.0384
2017	1.0355	1.0581
2018	1.0608	1.0782
2019	1.0899	1.0987
2020	1.1199	1.1196
2021	1.1507	1.1408
2022	1.1823	1.1625
2023	1.2148	1.1846
2024	1.2482	1.2071
2025	1.2825	1.2300
2026	1.3178	1.2534
2027	1.3541	1.2772
2028	1.3913	1.3015
2029	1.4296	1.3262
2030	1.4689	1.3514
2031	1.5094	1.3771
2032	1.5509	1.4033
2033	1.5936	1.4299
2034	1.6375	1.4571
2035	1.6826	1.4848
2036	1.7290	1.5130
2037	1.7766	1.5417
2038	1.8255	1.5710
2039	1.8758	1.6009
2040	1.9275	1.6313
2041	1.9807	1.6623
2042	2.0353	1.6939
2043	2.0914	1.7260
2044	2.1491	1.7588
2045	2.2083	1.7923
2046	2.2692	1.8263
2047	2.3318	1.8610
2048	2.3962	1.8964
2049	2.4623	1.9324
2050	2.5302	1.9691
2051	2.6001	2.0065
2052	2.6718	2.0447
2053	2.7456	2.0835
2054	2.8214	2.1231
2055	2.8993	2.1634
2056	2.9794	2.2045
2057	3.0616	2.2464
2058	3.1462	2.2891
2059	3.2331	2.3326
2060	3.3224	2.3769

Year	BUDGET YEAR INFLATION INDEX RESERVE (RE, RO, & RWO)	
	PA-RES	OM-RES
2014	1.0000	1.0000
2015	1.0105	1.0190
2016	1.0211	1.0384
2017	1.0355	1.0581
2018	1.0608	1.0782
2019	1.0899	1.0987
2020	1.1199	1.1196
2021	1.1507	1.1408
2022	1.1823	1.1625
2023	1.2148	1.1846
2024	1.2482	1.2071
2025	1.2825	1.2300
2026	1.3178	1.2534
2027	1.3540	1.2772
2028	1.3913	1.3015
2029	1.4296	1.3262
2030	1.4689	1.3514
2031	1.5093	1.3771
2032	1.5509	1.4033
2033	1.5936	1.4299
2034	1.6375	1.4571
2035	1.6826	1.4848
2036	1.7289	1.5130
2037	1.7765	1.5417
2038	1.8255	1.5710
2039	1.8758	1.6009
2040	1.9275	1.6313
2041	1.9806	1.6623
2042	2.0352	1.6939
2043	2.0913	1.7260
2044	2.1490	1.7588
2045	2.2082	1.7923
2046	2.2691	1.8263
2047	2.3317	1.8610
2048	2.3961	1.8964
2049	2.4622	1.9324
2050	2.5301	1.9691
2051	2.5999	2.0065
2052	2.6717	2.0447
2053	2.7454	2.0835
2054	2.8212	2.1231
2055	2.8991	2.1634
2056	2.9792	2.2045
2057	3.0615	2.2464
2058	3.1460	2.2891
2059	3.2329	2.3326
2060	3.3223	2.3769

Year	BUDGET YEAR INFLATION INDEX CIVILIAN (GS, WG, WL, & WS, SES)		
	Army CivPay	Army OMA-CIV	DoD-OMA
2014	1.0000	1.0000	1.0000
2015	1.0063	1.0190	1.0190
2016	1.0151	1.0384	1.0384
2017	1.0290	1.0581	1.0581
2018	1.0444	1.0782	1.0782
2019	1.0601	1.0987	1.0987
2020	1.0760	1.1196	1.1196
2021	1.0922	1.1408	1.1408
2022	1.1085	1.1625	1.1625
2023	1.1252	1.1846	1.1846
2024	1.1420	1.2071	1.2071
2025	1.1592	1.2300	1.2300
2026	1.1766	1.2534	1.2534
2027	1.1942	1.2772	1.2772
2028	1.2121	1.3015	1.3015
2029	1.2303	1.3262	1.3262
2030	1.2488	1.3514	1.3514
2031	1.2675	1.3771	1.3771
2032	1.2865	1.4033	1.4033
2033	1.3058	1.4299	1.4299
2034	1.3254	1.4571	1.4571
2035	1.3453	1.4848	1.4848
2036	1.3654	1.5130	1.5130
2037	1.3859	1.5417	1.5417
2038	1.4067	1.5710	1.5710
2039	1.4278	1.6009	1.6009
2040	1.4492	1.6313	1.6313
2041	1.4710	1.6623	1.6623
2042	1.4930	1.6939	1.6939
2043	1.5154	1.7260	1.7260
2044	1.5382	1.7588	1.7588
2045	1.5612	1.7923	1.7923
2046	1.5847	1.8263	1.8263
2047	1.6084	1.8610	1.8610
2048	1.6326	1.8964	1.8964
2049	1.6570	1.9324	1.9324
2050	1.6819	1.9691	1.9691
2051	1.7071	2.0065	2.0065
2052	1.7327	2.0447	2.0447
2053	1.7587	2.0835	2.0835
2054	1.7851	2.1231	2.1231
2055	1.8119	2.1634	2.1634
2056	1.8391	2.2045	2.2045
2057	1.8666	2.2464	2.2464
2058	1.8946	2.2891	2.2891
2059	1.9231	2.3326	2.3326
2060	1.9519	2.3769	2.3769

Year	BUDGET YEAR INFLATION INDEX Private Labor Market (PLM)
	Army OMA
2014	1.0000
2015	1.0190
2016	1.0384
2017	1.0581
2018	1.0782
2019	1.0987
2020	1.1196
2021	1.1408
2022	1.1625
2023	1.1846
2024	1.2071
2025	1.2300
2026	1.2534
2027	1.2772
2028	1.3015
2029	1.3262
2030	1.3514
2031	1.3771
2032	1.4033
2033	1.4299
2034	1.4571
2035	1.4848
2036	1.5130
2037	1.5417
2038	1.5710
2039	1.6009
2040	1.6313
2041	1.6623
2042	1.6939
2043	1.7260
2044	1.7588
2045	1.7923
2046	1.8263
2047	1.8610
2048	1.8964
2049	1.9324
2050	1.9691
2051	2.0065
2052	2.0447
2053	2.0835
2054	2.1231
2055	2.1634
2056	2.2045
2057	2.2464
2058	2.2891
2059	2.3326
2060	2.3769

Attachment 2

DEFAULT Summary Inflation Factor to Cost Element Mappings

ACTIVE DUTY PAY PLANS					
Pay Plan	LITE/PM Sort Seq	APPN	Category	Element	JIC INFLATION FACTOR MAPPING
ACTIVE ENLISTED					
AE	AE-1	ARMY MPA	Military Compensation	Avg Cost of Base Pay (Military)	Army MPA
AE	AE-2	ARMY MPA	Military Compensation	Avg Cost of Basic Allowance for Housing (In cash)	Army MPA
AE	AE-3	ARMY MPA	Military Compensation	Avg Cost of Basic Allowance for Subsistence	Army MPA
AE	AE-4	ARMY MPA	Other Benefits	Avg Cost of Other Benefits	Army MPA
AE	AE-5	ARMY MPA	Recruiting Costs	Avg Recruiting Cost for MOS (Amortized)	Army MPA
AE	AE-6	ARMY MPA	Retired Pay Accrual	Avg Cost of Retired Pay Accrual	Army MPA
AE	AE-7	ARMY MPA	Selective Reenlistment Bonus	Avg Cost of Reenlistment Bonus (A and B Amortized)	Army MPA
AE	AE-8	ARMY MPA	Separation Costs	Avg Cost of All Separation Incentives	Army MPA
AE	AE-9	ARMY MPA	Special Pays	Avg Cost of Special Pays	Not Inflated in PM
AE	AE-10	ARMY MPA	Training	Avg Cost of Training (Total Amortized)	Army MPA
AE	AE-11	ARMY MPA Non-Pay	Permanent Change of Station Costs	Avg Permanent Change of Station-Annualized	Army MPA Non-Pay
AE	AE-12	ARMY OMA	Morale, Welfare and Recreation Costs	Avg Cost of Morale, Welfare and Recreation	Army OMA
AE	AE-13	ARMY OMA	Recruiting Costs	Avg Recruiting Cost for MOS (Amortized)	Army OMA
AE	AE-14	ARMY OMA	Training	Avg Cost of Training (Total Amortized)	Army OMA
AE	AE-15	ARMY OMA_1	Montgomery GI Bill - Active Duty Costs	Avg Cost of MGB-AD Army College Fund (ACF)	Not Inflated in PM
AE	AE-16	ARMY OMA_1	Training	Avg Cost of Training (Total Amortized)	Army OMA
AE	AE-17	DoD OMA	Medical Support Costs	Avg Cost of Medical Support Cost	DoD OMA
AE	AE-18	DoD OMA	Other Benefits	Child Education (Impact Aid)	DoD OMA
AE	AE-19	DoD OMA	Other Benefits	Discount Groceries	DoD OMA
AE	AE-20	DoD OMA	Other Benefits	DoDEA and Family Assistance	DoD OMA
AE	AE-21	DoD OMA	Retirement	Treasury Contribution for Concurrent Receipts	DoD OMA
AE	AE-22	DoD OMA	Retirement	Treasury Contribution to Medicare Eligible Retiree Health Care Fund (MERHCF)	DoD OMA
AE	AE-23	DoD OMA	Retirement	Treasury Contribution to Retirement	DoD OMA
AE	AE-24	DoD OMA	Veteran Benefits	Veterans' Benefits (Cash and In-kind)	DoD OMA
ACTIVE OFFICER					
AO	AO-1	ARMY MPA	Military Compensation	Avg Cost of Base Pay (Military)	Army MPA
AO	AO-2	ARMY MPA	Military Compensation	Avg Cost of Basic Allowance for Housing (In cash)	Army MPA
AO	AO-3	ARMY MPA	Military Compensation	Avg Cost of Basic Allowance for Subsistence	Army MPA
AO	AO-4	ARMY MPA	Officer Acquisition Costs	Avg Cost of Officer Acquisition (Amortized)	Army MPA
AO	AO-5	ARMY MPA	Other Benefits	Avg Cost of Other Benefits	Army MPA
AO	AO-6	ARMY MPA	Retired Pay Accrual	Avg Cost of Retired Pay Accrual	Army MPA
AO	AO-7	ARMY MPA	Separation Costs	Avg Cost of All Separation Incentives	Army MPA
AO	AO-8	ARMY MPA	Special Pays	Avg Cost of Special Pays	Not Inflated in PM
AO	AO-9	ARMY MPA	Training	Avg Cost of Training (Amortized)	Army MPA
AO	AO-10	ARMY MPA Non-Pay	Permanent Change of Station Costs	Avg Permanent Change of Station-Annualized	Army MPA Non-Pay
AO	AO-11	ARMY OMA	Morale, Welfare and Recreation Costs	Avg Cost of Morale, Welfare and Recreation	Army OMA
AO	AO-12	ARMY OMA	Officer Acquisition Costs	Avg Cost of Officer Acquisition (Amortized)	Army OMA
AO	AO-13	ARMY OMA	Training	Avg Cost of Training (Amortized)	Army OMA
AO	AO-14	ARMY OMA_1	Training	Avg Cost of Training (Amortized)	Army OMA
AO	AO-15	DoD OMA	Medical Support Costs	Avg Cost of Medical Support Cost	DoD OMA
AO	AO-16	DoD OMA	Other Benefits	Child Education (Impact Aid)	DoD OMA
AO	AO-17	DoD OMA	Other Benefits	Discount Groceries	DoD OMA
AO	AO-18	DoD OMA	Other Benefits	DoDEA and Family Assistance	DoD OMA
AO	AO-19	DoD OMA	Retirement	Treasury Contribution for Concurrent Receipts	DoD OMA
AO	AO-20	DoD OMA	Retirement	Treasury Contribution to Medicare Eligible Retiree Health Care Fund (MERHCF)	DoD OMA
AO	AO-21	DoD OMA	Retirement	Treasury Contribution to Retirement	DoD OMA
AO	AO-22	DoD OMA	Veteran Benefits	Veterans' Benefits (Cash and In-kind)	DoD OMA
ACTIVE WARRANT OFFICER					
AWO	AWO-1	ARMY MPA	Military Compensation	Avg Cost of Base Pay (Military)	Army MPA
AWO	AWO-2	ARMY MPA	Military Compensation	Avg Cost of Basic Allowance for Housing (In cash)	Army MPA
AWO	AWO-3	ARMY MPA	Military Compensation	Avg Cost of Basic Allowance for Subsistence	Army MPA
AWO	AWO-4	ARMY MPA	Officer Acquisition Costs	Avg Cost of Officer Acquisition (Amortized)	Army MPA
AWO	AWO-5	ARMY MPA	Other Benefits	Avg Cost of Other Benefits	Army MPA
AWO	AWO-6	ARMY MPA	Retired Pay Accrual	Avg Cost of Retired Pay Accrual	Army MPA
AWO	AWO-7	ARMY MPA	Separation Costs	Avg Cost of All Separation Incentives	Army MPA
AWO	AWO-8	ARMY MPA	Special Pays	Avg Cost of Special Pays	Not Inflated in PM
AWO	AWO-9	ARMY MPA	Training	Avg Cost of Training (Amortized)	Army MPA
AWO	AWO-10	ARMY MPA Non-Pay	Permanent Change of Station Costs	Avg Permanent Change of Station-Annualized	Army MPA Non-Pay
AWO	AWO-11	ARMY OMA	Morale, Welfare and Recreation Costs	Avg Cost of Morale, Welfare and Recreation	Army OMA
AWO	AWO-12	ARMY OMA	Officer Acquisition Costs	Avg Cost of Officer Acquisition (Amortized)	Army OMA
AWO	AWO-13	ARMY OMA	Training	Avg Cost of Training (Amortized)	Army OMA
AWO	AWO-14	ARMY OMA_1	Officer Acquisition Costs	Avg Cost of Officer Acquisition (Amortized)	Army OMA
AWO	AWO-15	ARMY OMA_1	Training	Avg Cost of Training (Amortized)	Army OMA
AWO	AWO-16	DoD OMA	Medical Support Costs	Avg Cost of Medical Support Cost	DoD OMA
AWO	AWO-17	DoD OMA	Other Benefits	Child Education (Impact Aid)	DoD OMA
AWO	AWO-18	DoD OMA	Other Benefits	Discount Groceries	DoD OMA
AWO	AWO-19	DoD OMA	Other Benefits	DoDEA and Family Assistance	DoD OMA
AWO	AWO-20	DoD OMA	Retirement	Treasury Contribution for Concurrent Receipts	DoD OMA
AWO	AWO-21	DoD OMA	Retirement	Treasury Contribution to Medicare Eligible Retiree Health Care Fund (MERHCF)	DoD OMA
AWO	AWO-22	DoD OMA	Retirement	Treasury Contribution to Retirement	DoD OMA
AWO	AWO-23	DoD OMA	Veteran Benefits	Veterans' Benefits (Cash and In-kind)	DoD OMA

NATIONAL GUARD PAY PLANS					
Pay Plan	LITE/PM Sort Seq	APPN	Category	Element	JIC INFLATION FACTOR MAPPING
NG ENLISTED					
NE	NE-1	NG PA	Military Compensation	Avg Annualized Cost of Base Pay (Military)	NG-PA
NE	NE-2	NG PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Housing (in cash)	NG-PA
NE	NE-3	NG PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Subsistence	NG-PA
NE	NE-4	NG PA	Other Benefits	Avg Annualized Cost of FICA	NG-PA
NE	NE-5	NG PA	Other Benefits	Avg Fixed Cost of Clothing Allowance	NG-PA
NE	NE-6	NG PA	Other Benefits	Avg Fixed Cost of GI Bill	NG-PA
NE	NE-7	NG PA	Other Benefits	Avg Fixed Cost of Student Loan Repayment Program	NG-PA
NE	NE-8	NG PA	Other Benefits	Disability, Hospitalization & Death Gratuities	NG-PA
NE	NE-9	NG PA	Recruiting Costs	Avg Fixed Recruiting Cost for MOS (Amortized)	NG-PA
NE	NE-10	NG PA	Retired Pay Accrual	Avg Annualized Cost of Retired Pay Accrual	NG-PA
NE	NE-11	NG PA	Training Costs	Avg Fixed Cost of Training (Amortized)	NG-PA
NE	NE-12	NG OM	Medical Support Costs	Avg Annualized Cost of Health Care	NG-OM
NE	NE-13	NG OM	Recruiting Costs	Avg Fixed Recruiting Cost for MOS (Amortized)	NG-OM
NE	NE-14	NG OM	Training Costs	Avg Fixed Cost of Training (Amortized)	NG-OM
NE	NE-15	NG OM_1	Training Costs	Avg Cost of Training (Amortized)	NG-OM
NG OFFICER					
NO	NO-1	NG PA	Military Compensation	Avg Annualized Cost of Base Pay (Military)	NG-PA
NO	NO-2	NG PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Housing (in cash)	NG-PA
NO	NO-3	NG PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Subsistence	NG-PA
NO	NO-4	NG PA	Officer Acquisition	Avg Fixed Cost of Commissioning an Officer (Amortized)	NG-PA
NO	NO-5	NG PA	Other Benefits	Avg Annualized Cost of FICA	NG-PA
NO	NO-6	NG PA	Other Benefits	Avg Fixed Cost of Clothing Allowance	NG-PA
NO	NO-7	NG PA	Other Benefits	Disability, Hospitalization & Death Gratuities	NG-PA
NO	NO-8	NG PA	Retired Pay Accrual	Avg Annualized Cost of Retired Pay Accrual	NG-PA
NO	NO-9	NG PA	Training Costs	Avg Fixed Training Cost (Amortized)	NG-PA
NO	NO-10	NG OM	Other Benefits	Avg Annualized Cost of Health Care	NG-OM
NO	NO-11	NG OM	Training Costs	Avg Fixed Training Cost (Amortized)	NG-OM
NO	NO-12	NG OM_1	Training Costs	Avg Training Cost (Amortized)	NG-OM
NG WARRANT OFFICER					
NWO	NWO-1	NG PA	Military Compensation	Avg Annualized Cost of Base Pay (Military)	NG-PA
NWO	NWO-2	NG PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Housing (in cash)	NG-PA
NWO	NWO-3	NG PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Subsistence	NG-PA
NWO	NWO-4	NG PA	Other Benefits	Avg Annualized Cost of FICA	NG-PA
NWO	NWO-5	NG PA	Other Benefits	Avg Fixed Cost of Clothing Allowance	NG-PA
NWO	NWO-6	NG PA	Other Benefits	Disability, Hospitalization & Death Gratuities	NG-PA
NWO	NWO-7	NG PA	Retired Pay Accrual	Avg Annualized Cost of Retired Pay Accrual	NG-PA
NWO	NWO-8	NG PA	Training Costs	Avg Fixed Training Cost (Amortized)	NG-PA
NWO	NWO-9	NG OM	Other Benefits	Avg Annualized Cost of Health Care	NG-OM
NWO	NWO-10	NG OM	Training Costs	Avg Fixed Training Cost (Amortized)	NG-OM
NWO	NWO-11	NG OM_1	Training Costs	Avg Training Cost (Amortized)	NG-OM

RESERVE PAY PLANS					
Pay Plan	LITE/PM Sort Seq	APPN	Category	Element	JIC INFLATION FACTOR MAPPING
RESERVE ENLISTED					
RE	RE-1	RES PA	Military Compensation	Avg Annualized Cost of Base Pay (Military)	Res-PA
RE	RE-2	RES PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Housing (in cash)	Res-PA
RE	RE-3	RES PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Subsistence	Res-PA
RE	RE-4	RES PA	Other Benefits	Avg Annualized Cost of FICA	Res-PA
RE	RE-5	RES PA	Other Benefits	Avg Fixed Cost of Clothing Allowance	Res-PA
RE	RE-6	RES PA	Other Benefits	Avg Fixed Cost of GI Bill	Res-PA
RE	RE-7	RES PA	Other Benefits	Avg Fixed Cost of Student Loan Repayment Program	Res-PA
RE	RE-8	RES PA	Other Benefits	Disability, Hospitalization & Death Gratuities	Res-PA
RE	RE-9	RES PA	Recruiting Costs	Avg Fixed Recruiting Cost for MOS (Amortized)	Res-PA
RE	RE-10	RES PA	Retired Pay Accrual	Avg Annualized Cost of Retired Pay Accrual	Res-PA
RE	RE-11	RES PA	Training Costs	Avg Fixed Cost of Training (Amortized)	Res-PA
RE	RE-12	RES OM	Medical Support Costs	Avg Annualized Cost of Health Care	Res-OM
RE	RE-13	RES OM	Recruiting Costs	Avg Fixed Recruiting Cost for MOS (Amortized)	Res-OM
RE	RE-14	RES OM	Training Costs	Avg Fixed Cost of Training (Amortized)	Res-OM
RE	RE-15	RES OM_1	Training Costs	Avg Cost of Training (Amortized)	Res-OM
RESERVE OFFICER					
RO	RO-1	RES PA	Military Compensation	Avg Annualized Cost of Base Pay (Military)	Res-PA
RO	RO-2	RES PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Housing (in cash)	Res-PA
RO	RO-3	RES PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Subsistence	Res-PA
RO	RO-4	RES PA	Officer Acquisition	Avg Fixed Cost of Commissioning an Officer (Amortized)	Res-PA
RO	RO-5	RES PA	Other Benefits	Avg Annualized Cost of FICA	Res-PA
RO	RO-6	RES PA	Other Benefits	Avg Fixed Cost of Clothing Allowance	Res-PA
RO	RO-7	RES PA	Other Benefits	Disability, Hospitalization & Death Gratuities	Res-PA
RO	RO-8	RES PA	Retired Pay Accrual	Avg Annualized Cost of Retired Pay Accrual	Res-PA
RO	RO-9	RES PA	Training Costs	Avg Fixed Training Cost (Amortized)	Res-PA
RO	RO-10	RES OM	Other Benefits	Avg Annualized Cost of Health Care	Res-OM
RO	RO-11	RES OM	Training Costs	Avg Fixed Training Cost (Amortized)	Res-OM
RO	RO-12	RES OM_1	Training Costs	Avg Training Cost (Amortized)	Res-OM
RESERVE WARRANT OFFICER					
RWO	RWO-1	RES PA	Military Compensation	Avg Annualized Cost of Base Pay (Military)	Res-PA
RWO	RWO-2	RES PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Housing (in cash)	Res-PA
RWO	RWO-3	RES PA	Military Compensation	Avg Annualized Cost of Basic Allowance for Subsistence	Res-PA
RWO	RWO-4	RES PA	Other Benefits	Avg Annualized Cost of FICA	Res-PA
RWO	RWO-5	RES PA	Other Benefits	Avg Fixed Cost of Clothing Allowance	Res-PA
RWO	RWO-6	RES PA	Other Benefits	Disability, Hospitalization & Death Gratuities	Res-PA
RWO	RWO-7	RES PA	Retired Pay Accrual	Avg Annualized Cost of Retired Pay Accrual	Res-PA
RWO	RWO-8	RES PA	Training Costs	Avg Fixed Training Cost (Amortized)	Res-PA
RWO	RWO-9	RES OM	Other Benefits	Avg Annualized Cost of Health Care	Res-OM
RWO	RWO-10	RES OM	Training Costs	Avg Fixed Training Cost (Amortized)	Res-OM
RWO	RWO-11	RES OM_1	Training Costs	Avg Training Cost (Amortized)	Res-OM

CIVILIAN PAY PLANS					
Pay Plan	LITE/PM Sort Seq	APPN	Category	Element	JIC INFLATION FACTOR MAPPING
CIVILIAN GENERAL SCHEDULE					
GS	GS-1	ARMY CivPay	Compensation	Avg Cost of Base Pay (Civilian)	Army CIVPAY
GS	GS-2	ARMY CivPay	Compensation	Avg Cost of Holiday Pay	Army CIVPAY
GS	GS-3	ARMY CivPay	Compensation	Avg Cost of Other Compensation	Army CIVPAY
GS	GS-4	ARMY CivPay	Compensation	Avg Cost of Overtime Pay	Army CIVPAY
GS	GS-5	ARMY CivPay	Other Benefits	Avg Cost of Benefits	Army CIVPAY
GS	GS-6	ARMY CivPay	Other Benefits	Avg Cost of Cash Awards	Army CIVPAY
GS	GS-7	ARMY CivPay	Other Benefits	Avg Cost of Former Employee Compensation	Army CIVPAY
GS	GS-8	ARMY OMA	Training	Training	Army OMA
GS	GS-9	DoD OMA	Retired Pay Accrual	Government-Funded Retirement	DoD OMA
CIVILIAN SENIOR EXECUTIVE SERVICE SCHEDULE					
SES	SES-1	ARMY CivPay	Compensation	Avg Cost of Base Pay (Civilian)	Army CIVPAY
SES	SES-2	ARMY CivPay	Other Benefits	Avg Cost of Federal Employees Gov't Health Insurance	Army CIVPAY
SES	SES-3	ARMY CivPay	Other Benefits	Avg Cost of Federal Employees Gov't Life Insurance	Army CIVPAY
SES	SES-4	ARMY CivPay	Retired Pay Accrual	Avg Cost of Army-Funded Retirement	Army CIVPAY
SES	SES-5	ARMY OMA	Training	Training	Army OMA
SES	SES-6	DoD OMA	Retired Pay Accrual	Government-Funded Retirement	DoD OMA
CIVILIAN WAGE GRADE					
WG	WG-1	ARMY CivPay	Compensation	Avg Cost of Base Pay (Civilian)	Army CIVPAY
WG	WG-2	ARMY CivPay	Compensation	Avg Cost of Premium Pay	Army CIVPAY
WG	WG-3	ARMY CivPay	Other Benefits	Avg Cost of Federal Employees Gov't Health Insurance	Army CIVPAY
WG	WG-4	ARMY CivPay	Other Benefits	Avg Cost of Federal Employees Gov't Life Insurance	Army CIVPAY
WG	WG-5	ARMY CivPay	Other Benefits	Avg Cost of Miscellaneous Pay	Army CIVPAY
WG	WG-6	ARMY CivPay	Retired Pay Accrual	Avg Cost of Army-Funded Retirement	Army CIVPAY
WG	WG-7	ARMY OMA	Training	Training	Army OMA
WG	WG-8	DoD OMA	Retired Pay Accrual	Government-Funded Retirement	DoD OMA
CIVILIAN WAGE LEADER					
WL	WL-1	ARMY CivPay	Compensation	Avg Cost of Base Pay (Civilian)	Army CIVPAY
WL	WL-2	ARMY CivPay	Compensation	Avg Cost of Premium Pay	Army CIVPAY
WL	WL-3	ARMY CivPay	Other Benefits	Avg Cost of Federal Employees Gov't Health Insurance	Army CIVPAY
WL	WL-4	ARMY CivPay	Other Benefits	Avg Cost of Federal Employees Gov't Life Insurance	Army CIVPAY
WL	WL-5	ARMY CivPay	Other Benefits	Avg Cost of Miscellaneous Pay	Army CIVPAY
WL	WL-6	ARMY CivPay	Retired Pay Accrual	Avg Cost of Army-Funded Retirement	Army CIVPAY
WL	WL-7	ARMY OMA	Training	Training	Army OMA
WL	WL-8	DoD OMA	Retired Pay Accrual	Government-Funded Retirement	DoD OMA
CIVILIAN WAGE SUPERVISOR					
WS	WS-1	ARMY CivPay	Compensation	Avg Cost of Base Pay (Civilian)	Army CIVPAY
WS	WS-2	ARMY CivPay	Compensation	Avg Cost of Premium Pay	Army CIVPAY
WS	WS-3	ARMY CivPay	Other Benefits	Avg Cost of Federal Employees Gov't Health Insurance	Army CIVPAY
WS	WS-4	ARMY CivPay	Other Benefits	Avg Cost of Federal Employees Gov't Life Insurance	Army CIVPAY
WS	WS-5	ARMY CivPay	Other Benefits	Avg Cost of Miscellaneous Pay	Army CIVPAY
WS	WS-6	ARMY CivPay	Retired Pay Accrual	Avg Cost of Army-Funded Retirement	Army CIVPAY
WS	WS-7	ARMY OMA	Training	Training	Army OMA
WS	WS-8	DoD OMA	Retired Pay Accrual	Government-Funded Retirement	DoD OMA
PRIVATE LABOR MARKET (PLM)					
PLM	PLM-1	Private Sector Funded	Compensation	Avg Cost of Salary	Army OMA
PLM	PLM-2	Private Sector Funded	Compensation	Avg Cost of Benefits	Army OMA
PLM	PLM-3	Private Sector Funded	Overhead	Overhead	Army OMA